

Suman Travels — Product Requirements Document (PRD)

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1. Executive Summary

Suman Travels is a premium chauffeur-driven car rental platform based in Mumbai, offering airport transfers, local rentals, outstation trips, and corporate travel. The platform features a dark, luxury-themed UI with glassmorphism effects, coral/orange accents, and a Mumbai-centric brand identity. The system supports online booking, fleet browsing, pricing transparency, and customer account management.

Target Audience: Business travelers, tourists, corporate clients, and local residents seeking premium and standard vehicle rentals in Mumbai and surrounding areas (Pune, Lonavala, Nashik, etc.).

2. Brand Identity & Design System

2.1 Color Palette

Token	Hex Value	Usage
--color-bg-primary	#0B0F19	Main page background (deep navy-black)
--color-bg-secondary	#111827	Card backgrounds, sections
--color-bg-glass	rgba(17, 24, 39, 0.65)	Glassmorphism cards
--color-bg-glass-border	rgba(255, 255, 255, 0.08)	Glass card borders
--color-accent-primary	#FF6B47	Primary buttons, CTAs, highlights, links
--color-accent-hover	#E85A38	Button hover states
--color-accent-gradient-start	#FF8A65	Gradient accents (hero text)
--color-accent-gradient-end	#FF6B47	Gradient accents (hero text)
--color-text-primary	FFFFFF	Headings, primary text
--color-text-secondary	#9CA3AF	Body text, descriptions
--color-text-muted	#6B7280	Labels, placeholders
--color-success	#10B981	Success states, confirmations
--color-warning	#F59E0B	Warning alerts
--color-error	#EF4444	Error states, validation
--color-overlay	rgba(0, 0, 0, 0.4)	Image overlays, backdrops

2.2 Typography

Element	Font Family	Weight	Size	Line Height	Letter Spacing
H1 (Hero)	Inter	800 (Extra Bold)	56px / 3.5rem	1.1	-0.02em
H2 (Section)	Inter	700 (Bold)	36px / 2.25rem	1.2	-0.01em
H3 (Card Title)	Inter	600 (Semi Bold)	20px / 1.25rem	1.3	0
Body	Inter	400 (Regular)	16px / 1rem	1.6	0
Body Small	Inter	400 (Regular)	14px / 0.875rem	1.5	0
Caption/Label	Inter	500 (Medium)	12px / 0.75rem	1.4	0.05em
Button	Inter	600 (Semi Bold)	14px / 0.875rem	1	0.02em
Nav Link	Inter	500 (Medium)	14px / 0.875rem	1	0
Price	Inter	700 (Bold)	24px / 1.5rem	1.2	-0.01em

Font Import:

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
  rel="stylesheet">
```

2.3 Spacing & Layout

- **Container Max Width:** 1280px (centered)
- **Section Padding:** py-20 (80px vertical) on desktop, py-12 (48px) on mobile
- **Card Padding:** 24px (p-6)
- **Card Border Radius:** 16px (rounded-2xl)
- **Button Border Radius:** 9999px (rounded-full) for primary CTAs, 8px (rounded-lg) for secondary
- **Grid Gap:** 24px (gap-6)
- **Glassmorphism Effect:**
 - backdrop-filter: blur(16px)
 - background: rgba(17, 24, 39, 0.65)
 - border: 1px solid rgba(255, 255, 255, 0.08)
 - box-shadow: 0 4px 30px rgba(0, 0, 0, 0.1)

2.4 Component Styles

Primary Button (Book Now / Search):

- Background: #FF6B47
- Text: #FFFFFF

- Padding: 12px 32px
- Border Radius: 9999px (pill shape)
- Hover: Background #E85A38, transform translateY(-2px), box-shadow 0 10px 25px rgba(255, 107, 71, 0.3)
- Transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1)

Secondary Button (Call Now / Outline):

- Background: transparent
- Border: 1px solid rgba(255, 255, 255, 0.2)
- Text: #FFFFFF
- Hover: Background rgba(255, 255, 255, 0.1)

Service Card:

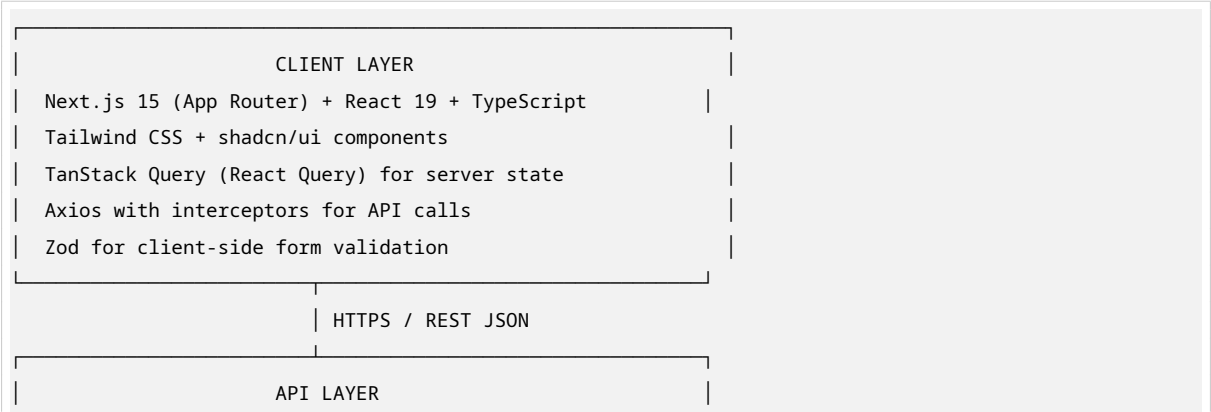
- Background: rgba(17, 24, 39, 0.8)
- Border: 1px solid rgba(255, 255, 255, 0.05)
- Border Radius: 16px
- Hover: Border color rgba(255, 107, 71, 0.3), transform translateY(-4px)
- Icon Container: 48px circle, background rgba(255, 107, 71, 0.15), icon color #FF6B47

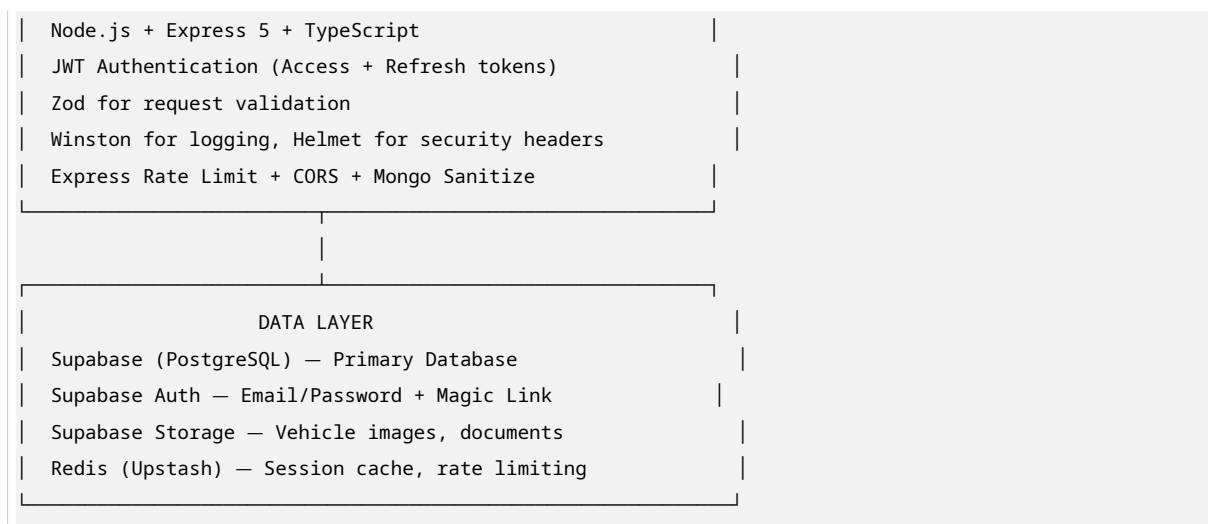
Input Field (Booking Form):

- Background: rgba(255, 255, 255, 0.05)
- Border: 1px solid rgba(255, 255, 255, 0.1)
- Border Radius: 12px
- Text: #FFFFFF
- Placeholder: #6B7280
- Focus: Border color #FF6B47, ring 0 0 0 3px rgba(255, 107, 71, 0.2)

3. Tech Stack

3.1 Architecture Overview





3.2 Detailed Stack

Layer	Technology	Version	Purpose
Frontend Framework	Next.js	15.x	App Router, SSR/SSG, API routes (optional)
UI Library	React	19.x	Component-based UI
Language	TypeScript	5.6+	Type safety across stack
Styling	Tailwind CSS	4.x	Utility-first CSS
UI Components	shadcn/ui	latest	Accessible, customizable components
Animation	Framer Motion	latest	Page transitions, micro-interactions
Icons	Lucide React	latest	Consistent iconography
State Management	TanStack Query	5.x	Server state, caching, mutations
HTTP Client	Axios	latest	API communication with interceptors
Validation	Zod	latest	Schema validation (forms + API)
Backend Runtime	Node.js	20.x LTS	Server runtime
Backend Framework	Express	5.x	REST API framework
Database	Supabase (PostgreSQL)	—	Primary data store
Auth	Supabase Auth	—	Email/password, JWT handling
Storage	Supabase Storage	—	Image uploads
Cache	Redis (Upstash)	—	Sessions, rate limits
Email	Resend / Nodemailer	—	Booking confirmations, OTP
Payments	Razorpay / Stripe	—	Online payments (future)
Monitoring	Sentry	—	Error tracking

Table 3 – continued

Layer	Technology	Version	Purpose
Deployment	Vercel (Frontend) + Render/ Railway (Backend)	—	Production hosting

4. Database Schema (Supabase PostgreSQL)

4.1 Users Table (Managed by Supabase Auth)

```
-- Supabase Auth handles: id, email, encrypted_password, email_confirmed_at, etc.
-- Extended profile:
CREATE TABLE public.profiles (
  id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
  full_name VARCHAR(100) NOT NULL,
  phone VARCHAR(20) NOT NULL UNIQUE,
  avatar_url TEXT,
  role VARCHAR(20) DEFAULT 'customer' CHECK (role IN ('customer', 'admin', 'driver')),
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.2 Vehicles Table

```
CREATE TABLE public.vehicles (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(100) NOT NULL,           -- e.g., "Toyota Hycross"
  category VARCHAR(50) NOT NULL,        -- "suv", "sedan", "luxury", "muv"
  brand VARCHAR(50) NOT NULL,           -- "Toyota", "Maruti", "BMW"
  model VARCHAR(50) NOT NULL,           -- "Hycross", "Innova", "Crysta"
  seating_capacity INT NOT NULL DEFAULT 4,
  luggage_capacity INT,                 -- Number of bags
  features TEXT[],                      -- ['AC', 'GPS', 'Leather Seats', 'Wifi']
  image_url TEXT,
  is_active BOOLEAN DEFAULT true,
  is_luxury BOOLEAN DEFAULT false,
  display_order INT DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.3 Pricing Table

```
CREATE TABLE public.pricing (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  vehicle_id UUID REFERENCES public.vehicles(id) ON DELETE CASCADE,
  package_type VARCHAR(50) NOT NULL,    -- "local_8hrs", "local_12hrs", "airport", "outstation"
  duration_hours INT,                   -- NULL for airport/outstation
  distance_km INT,                      -- NULL for airport/outstation
);
```

```

base_price DECIMAL(10,2) NOT NULL,          -- In INR
extra_hour_rate DECIMAL(10,2),              -- Per extra hour
extra_km_rate DECIMAL(10,2),                -- Per extra km
night_charge DECIMAL(10,2) DEFAULT 0,       -- After 23:00
description TEXT,
is_active BOOLEAN DEFAULT true,
created_at TIMESTAMPTZ DEFAULT NOW()
);

```

4.4 Bookings Table

```

CREATE TABLE public.bookings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
  vehicle_id UUID REFERENCES public.vehicles(id),
  pricing_id UUID REFERENCES public.pricing(id),
  booking_type VARCHAR(50) NOT NULL,        -- "local", "airport", "outstation"
  status VARCHAR(30) DEFAULT 'pending'
    CHECK (status IN ('pending', 'confirmed', 'assigned', 'in_progress', 'completed', 'cancelled')),

  -- Trip Details
  pickup_location VARCHAR(255) NOT NULL,
  drop_location VARCHAR(255),
  pickup_date DATE NOT NULL,
  pickup_time TIME NOT NULL,
  return_date DATE,                        -- For round trips

  -- Passenger Details
  passenger_name VARCHAR(100) NOT NULL,
  passenger_phone VARCHAR(20) NOT NULL,
  passenger_email VARCHAR(100),

  -- Pricing Breakdown
  base_price DECIMAL(10,2) NOT NULL,
  extra_hours INT DEFAULT 0,
  extra_km INT DEFAULT 0,
  extra_charges DECIMAL(10,2) DEFAULT 0,
  night_charges DECIMAL(10,2) DEFAULT 0,
  total_amount DECIMAL(10,2) NOT NULL,

  -- Payment
  payment_status VARCHAR(30) DEFAULT 'pending'
    CHECK (payment_status IN ('pending', 'partial', 'paid', 'refunded')),
  payment_method VARCHAR(50),              -- "cash", "online", "upi"

  -- Driver Assignment
  driver_id UUID REFERENCES public.profiles(id),
  driver_notes TEXT,

  -- Metadata
  special_requests TEXT,

```

```
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.5 Drivers Table

```
CREATE TABLE public.drivers (
  id UUID PRIMARY KEY REFERENCES public.profiles(id) ON DELETE CASCADE,
  license_number VARCHAR(100) NOT NULL UNIQUE,
  license_expiry DATE NOT NULL,
  vehicle_assigned UUID REFERENCES public.vehicles(id),
  is_available BOOLEAN DEFAULT true,
  rating DECIMAL(2,1) DEFAULT 5.0 CHECK (rating >= 1 AND rating <= 5),
  total_trips INT DEFAULT 0,
  current_location GEOGRAPHY(POINT),      -- For live tracking (future)
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.6 Contact / Enquiries Table

```
CREATE TABLE public.enquiries (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) NOT NULL,
  phone VARCHAR(20),
  subject VARCHAR(100) NOT NULL,
  message TEXT NOT NULL,
  status VARCHAR(30) DEFAULT 'new' CHECK (status IN ('new', 'in_progress', 'resolved')),
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.7 Policies / CMS Table

```
CREATE TABLE public.pages (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  slug VARCHAR(100) UNIQUE NOT NULL,      -- "privacy-policy", "terms"
  title VARCHAR(200) NOT NULL,
  content TEXT NOT NULL,                  -- HTML/Markdown content
  meta_description TEXT,
  is_published BOOLEAN DEFAULT true,
  updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

5. API Specification (Node.js + Express)

5.1 Authentication Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/auth/register	No	Register with email, password, name, phone
POST	/api/v1/auth/login	No	Login, returns access token + sets refresh cookie
POST	/api/v1/auth/refresh	No	Refresh access token using HttpOnly cookie
POST	/api/v1/auth/logout	Yes	Clear refresh cookie, invalidate session
POST	/api/v1/auth/forgot-password	No	Send password reset email
POST	/api/v1/auth/reset-password	No	Reset password with token
GET	/api/v1/auth/me	Yes	Get current user profile
PATCH	/api/v1/auth/me	Yes	Update profile

5.2 Vehicle Endpoints

Method	Endpoint	Auth	Description
GET	/api/v1/vehicles	No	List all active vehicles with filters
GET	/api/v1/vehicles/:id	No	Get vehicle details with pricing
GET	/api/v1/vehicles/categories	No	List vehicle categories

5.3 Pricing Endpoints

Method	Endpoint	Auth	Description
GET	/api/v1/pricing	No	Get all pricing packages
GET	/api/v1/pricing/vehicle/:vehicleId	No	Get pricing for specific vehicle
POST	/api/v1/pricing/calculate	No	Calculate estimated fare

5.4 Booking Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/bookings	Yes	Create new booking
GET	/api/v1/bookings	Yes	List user' s bookings (customer) / all (admin)
GET	/api/v1/bookings/:id	Yes	Get booking details
PATCH	/api/v1/bookings/:id	Yes	Update booking (status, driver, etc.)
DELETE	/api/v1/bookings/:id	Yes	Cancel booking
POST	/api/v1/bookings/:id/confirm	Admin	Confirm booking

Table 7 – continued

Method	Endpoint	Auth	Description
POST	/api/v1/bookings/:id/ assign-driver	Admin	Assign driver

5.5 Contact / Enquiry Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/contact	No	Submit contact form
GET	/api/v1/contact	Admin	List all enquiries

5.6 Admin Endpoints

Method	Endpoint	Auth	Description
GET	/api/v1/admin/ dashboard	Admin	Dashboard stats
GET	/api/v1/admin/ bookings	Admin	All bookings with filters
GET	/api/v1/admin/drivers	Admin	Manage drivers
GET	/api/v1/admin/reports	Admin	Revenue reports

5.7 Standard Response Format

```
// Success
{
  "success": true,
  "data": { ... },
  "message": "Operation successful"
}

// Error
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid input data",
    "fields": {
      "email": ["Invalid email format"]
    }
  }
}
```

6. Page-by-Page Requirements

6.1 Home Page (/)

Layout Structure:

1. **Header/Navbar** (Fixed, glassmorphism on scroll)
 - Logo: “Suman Travels” with car icon
 - Nav Links: Fleet, Services, Offers, About Us, Contact
 - CTA: “Book Now” (pill button, orange)
 - Auth: Login / User avatar (if logged in)
2. **Hero Section**
 - Background: Full-width Mumbai cityscape/sunset image with dark gradient overlay
 - Tagline Badge: “Skip the traffic, own the road.” (with icon)
 - Headline: “Drive Mumbai.” (white) + “Live the Journey.” (orange gradient)
 - Subheadline: “Premium cars, professional chauffeurs, and unmatched comfort — anytime, anywhere. From the airport to Marine Drive, experience seamless luxury.”
 - CTAs: “Book Your Ride” (primary, orange pill) + “Call Now” (secondary, outline)
 - **Quick Booking Widget** (Glassmorphism card):
 - Pick-up Location (dropdown: Mumbai areas)
 - Drop-off Location (dropdown: Mumbai, Pune, Lonavala, Nashik, Airport)
 - Date & Time Picker
 - Vehicle Type (dropdown: All, SUV, Sedan, Luxury)
 - “Search” Button (orange)
3. **Stats Bar** (4 columns, glass cards)
 - 5000+ Happy Customers
 - 50+ Premium Vehicles
 - 100+ Destinations
 - 24/7 Customer Support
4. **Our Premium Services** (4 cards)
 - Airport Transfers (plane icon)
 - Local Rentals (building icon)
 - Outstation Trips (map icon)
 - Corporate Travel (briefcase icon)
 - Each card: Icon, Title, Description, “Explore →” link
5. **Featured Fleet** (Horizontal scroll or grid)
 - Show top 4-6 vehicles
 - Card: Image, Name, Category, Starting Price, “View Details” button
6. **Why Choose Us** (3-4 value propositions)
 - Professional Drivers
 - Transparent Pricing
 - 24/7 Support
 - Clean & Sanitized Vehicles
7. **Testimonials** (Carousel)
 - 3-5 customer reviews with star ratings
8. **Footer**

- Logo + Tagline
- Quick Links: Home, Fleet, Services, About, Contact, Privacy Policy
- Contact Info: Phone, Email, Address (Mumbai)
- Social Links
- Copyright: “ 2024 Suman Travels Mumbai. Elite Chauffeur Services.”

Interactions:

- Navbar: Transparent → Glassmorphism on scroll (trigger at 50px)
- Hero text: Fade-in + translateY animation on load
- Stats: Count-up animation when in viewport
- Service cards: Hover lift effect with border glow
- Booking widget: Real-time validation

6.2 Header Component (Global)

- Fixed position, z-50
- Height: 72px
- Background: `rgba(11, 15, 25, 0.8)` + `backdrop-blur-xl` on scroll
- Logo: Left aligned, white text with small orange accent
- Navigation: Center-aligned (desktop), hamburger menu (mobile)
- Active link: Orange underline indicator
- Mobile menu: Full-screen overlay, dark background, slide-in from right

6.3 Fleet Page (/fleet)

Layout:

1. Page Header: “Our Premium Fleet” + subtitle
2. Filter Bar:
 - Category tabs: All, SUV, Sedan, MUV, Luxury
 - Sort: Price Low-High, Popular
3. Vehicle Grid (responsive: 1 col mobile, 2 col tablet, 3 col desktop):
 - Vehicle Card:
 - Image (aspect-ratio 16:9, object-cover)
 - Category badge (top-left)
 - Vehicle Name
 - Features list (seats, luggage, AC, etc.)
 - Starting Price: “From X,XXX”
 - “Book Now” button
 - “View Details” link

4. Vehicle Detail Modal/Page:

- Large image gallery
 - Full specifications
 - Pricing table (all packages)
 - “Book This Vehicle” CTA
-

6.4 Services Page (/services)

Layout:

1. Page Header: “Our Services”
 2. Service Cards (full-width, alternating layout):
 - **Airport Transfers:** Image left, content right. Details about pickup/drop, flight tracking, meet & greet.
 - **Local Rentals:** Content left, image right. Hourly/daily packages, Mumbai city coverage.
 - **Outstation Trips:** Image left, content right. Pune, Lonavala, Nashik, Shirdi, etc.
 - **Corporate Travel:** Content left, image right. B2B solutions, monthly billing, dedicated account manager.
 3. Each service section includes:
 - Icon + Title
 - Description
 - Key benefits (bullet points)
 - Starting price
 - “Book Now” button
-

6.5 About Page (/about)

Layout:

1. Hero: “About Suman Travels” with company image
 2. Our Story: Company history, mission, vision
 3. Why Choose Us: 4-6 value cards with icons
 4. Our Fleet Stats: Vehicle count, driver count, years of service
 5. Service Areas: Map or list of covered cities (Mumbai, Pune, Thane, Navi Mumbai, etc.)
 6. Team/Drivers: Optional driver profiles
-

6.6 Pricing Page (/pricing)

Layout:

1. Page Header: “Transparent Pricing”
2. Pricing Table (Responsive):
 - Vehicle Name | 8 Hrs / 80 Km | 12 Hrs / 100 Km | Extra Hr | Extra Km | Night Charge | Airport
 - Rows for each vehicle with actual prices
3. Pricing Cards (Mobile view):
 - One card per vehicle with all packages listed
4. Notes Section:
 - “Night charges apply after 23:00”
 - “Toll and parking charges extra”
 - “Prices inclusive of driver allowance”
 - “GST applicable”
5. CTA: “Get Custom Quote” button

Pricing Data (from client input):

Vehicle	8Hrs/80Km	12Hrs/100Km	Extra Hr	Extra Km	Night Charge	Airport P&D
Toyota Hycross	4,500	6,000	300	30	—	—
Toyota Innova	3,000	4,000	200	20	200	2,800
Toyota Crysta	3,800	5,000	250	23	200	2,200
Small Car (Dzire/ Etios)	2,500	3,200	200	15	200	1,500
Maruti Ertiga	3,000	4,000	200	18	200	2,000
Luxury (BMW, Audi, Mercedes)	On Request	On Request	On Request	On Request	On Request	On Request

6.7 Contact Page (/contact)

Layout:

1. Page Header: “Get in Touch”
2. Two-column layout:

- Left: Contact form
 - Name (required)
 - Email (required)
 - Phone (required)
 - Subject (dropdown: General, Booking, Corporate, Feedback)
 - Message (textarea)
 - “Send Message” button
 - Right: Contact information
 - Phone numbers (primary + secondary)
 - WhatsApp number (with click-to-chat)
 - Email address
 - Office address (Mumbai)
 - Operating hours: “24/7 Available”
 - Embedded map (Google Maps)
3. Success toast after form submission
-

6.8 Privacy & Policy Page (/privacy-policy)

- Static content page
 - Sections: Data Collection, Usage, Sharing, Cookies, Security, Your Rights, Contact
 - Dark theme consistent with site
 - Rendered from CMS (pages table) or static markdown
-

6.9 My Booking Page (/my-bookings) — Authenticated

Layout:

1. Page Header: “My Bookings”
2. Booking Tabs: Upcoming | Past | Cancelled
3. Booking Card:
 - Booking ID
 - Vehicle Name + Image
 - Trip: Pickup → Drop-off
 - Date & Time
 - Status badge (color-coded: pending=yellow, confirmed=green, completed=blue, cancelled=red)
 - Total Amount
 - Actions: View Details, Cancel (if pending), Download Invoice (if completed), Rebook

4. Empty state: “No bookings yet. Book your first ride!” with CTA
-

6.10 Booking Flow (/book)

Step-by-step wizard:

1. Step 1: Select Vehicle

- Grid of vehicles with pricing
- Filter by category
- Search by name

2. Step 2: Trip Details

- Pickup location (autocomplete)
- Drop location (autocomplete)
- Pickup date (calendar)
- Pickup time (time picker)
- Return date (for round trips)
- Package selection (8hrs/12hrs/Airport/Outstation)

3. Step 3: Passenger Details

- Name (pre-filled if logged in)
- Phone (pre-filled)
- Email (pre-filled)
- Special requests (textarea)

4. Step 4: Review & Confirm

- Booking summary card
- Price breakdown:
 - Base price
 - Extra hours (if any)
 - Extra KM (if any)
 - Night charges (if applicable)
 - Total
- Terms checkbox
- “Confirm Booking” button

5. Step 5: Confirmation

- Success animation
 - Booking ID
 - “View My Bookings” button
 - Share booking details option
-

7. Authentication & Authorization

7.1 Auth Flow

1. Registration:

- Email, Password, Full Name, Phone
- Email verification required (Supabase sends OTP/link)

- Auto-create profile in `profiles` table via trigger

2. Login:

- Email + Password
- JWT Access Token (15 min expiry) stored in memory
- Refresh Token (7 days) in `HttpOnly`, `Secure`, `SameSite=Strict` cookie
- Token refresh handled by Axios interceptor

3. Password Reset:

- “Forgot Password” link
- Email with reset token
- New password form

4. Session Management:

- Auto-refresh on token expiry
- Logout clears all tokens
- “Remember me” option extends refresh token

7.2 Role-Based Access

Role	Permissions
customer	View fleet, create bookings, view own bookings, edit profile
driver	View assigned bookings, update trip status, view route
admin	Full access: manage vehicles, pricing, all bookings, drivers, enquiries, reports, CMS

7.3 Route Guards

- Public routes: Home, Fleet, Services, About, Pricing, Contact, Privacy
- Auth-only routes: My Bookings, Profile, Book (stepper)
- Admin routes: `/admin/*` — redirect non-admin to 403 or home

8. Functional Requirements

8.1 Booking System

- Users can book without registration (guest booking) — optional
- Registered users see booking history
- Real-time availability check (vehicle not double-booked for same time)
- Booking confirmation email/SMS
- Admin notification on new booking
- Status updates: Pending → Confirmed → Assigned → In Progress → Completed
- Cancellation policy: Free cancellation up to 2 hours before pickup
- Automatic price calculation based on package + extras

8.2 Vehicle Management (Admin)

- CRUD operations for vehicles
- Upload vehicle images (Supabase Storage)
- Toggle vehicle availability

- Manage pricing packages per vehicle
- Vehicle categories: SUV, Sedan, MUV, Luxury

8.3 Driver Management (Admin)

- Driver registration and verification
- License tracking with expiry alerts
- Assign drivers to bookings
- Driver availability toggle
- Rating system

8.4 Communication

- Booking confirmation email
- Booking reminder (2 hours before)
- Driver assignment notification
- Contact form submission to admin email
- WhatsApp integration (future)

8.5 Search & Filter

- Vehicle search by name
 - Filter by category, price range, seating capacity
 - Sort by price, popularity, rating
 - Location-based suggestions
-

9. Non-Functional Requirements

9.1 Performance

- Page load time: < 2s (Lighthouse score > 90)
- Time to Interactive: < 3s on 3G
- Image optimization: WebP format, lazy loading, responsive images
- API response time: < 200ms (p95)

9.2 Security

- HTTPS only (HSTS enabled)
- Security headers via Helmet:
 - Content Security Policy (CSP)
 - X-Frame-Options: DENY
 - X-Content-Type-Options: nosniff
- Input sanitization (prevent XSS, NoSQL injection)
- Rate limiting: 100 req/min general, 10 req/min auth routes
- Password requirements: min 8 chars, 1 uppercase, 1 number, 1 special char
- SQL injection prevention via parameterized queries (Supabase)
- CORS: Whitelist only known origins

9.3 SEO

- Meta tags for all pages (title, description, OG tags)
- Structured data (JSON-LD) for LocalBusiness
- Sitemap.xml generation
- robots.txt
- Canonical URLs
- Semantic HTML

9.4 Accessibility

- WCAG 2.1 AA compliance
- Keyboard navigation support
- ARIA labels for interactive elements
- Color contrast ratio: 4.5:1 minimum
- Focus indicators visible
- Screen reader friendly

9.5 Responsive Breakpoints

Breakpoint	Width	Target
Mobile	< 640px	Phones
Tablet	640px - 1024px	Tablets
Desktop	1024px - 1280px	Laptops
Wide	> 1280px	Desktops

10. Data Flow & State Management

10.1 Booking Flow

```
User selects vehicle → Frontend validates → POST /api/v1/bookings
→ Backend validates (Zod) → Check vehicle availability → Calculate price
→ Create booking record (status: pending) → Send confirmation email
→ Return booking ID → Frontend redirects to confirmation page
→ Admin dashboard shows new booking → Admin confirms/assigns driver
→ User receives status update email
```

10.2 Auth Flow

```
User login → POST /api/v1/auth/login → Validate credentials
→ Generate access token (15min) + refresh token (7days)
→ Set refresh token as HttpOnly cookie
→ Return access token + user data → Frontend stores in memory
→ Axios interceptor attaches token to requests
→ On 401, call /auth/refresh → Update access token → Retry request
→ On refresh failure → Redirect to login
```

11. Third-Party Integrations

Service	Purpose	Environment Variables
Supabase	Database, Auth, Storage	SUPABASE_URL, SUPABASE_ANON_KEY, SUPABASE_SERVICE_ROLE_KEY
Resend / SendGrid	Transactional emails	SMTP_HOST, SMTP_USER, SMTP_PASS
Google Maps	Location autocomplete, embedded maps	GOOGLE_MAPS_API_KEY
Razorpay	Payment processing (future)	RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET
Sentry	Error monitoring	SENTRY_DSN
Redis (Upstash)	Session cache, rate limiting	REDIS_URL, REDIS_TOKEN

12. Environment Variables

12.1 Frontend (.env.local)

```
# API
NEXT_PUBLIC_API_BASE_URL=http://localhost:5000

# Supabase
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=your-anon-key

# Google Maps
NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=your-maps-key

# App Config
NEXT_PUBLIC_APP_NAME=Suman Travels
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

12.2 Backend (.env)

```
# Server
NODE_ENV=development
PORT=5000
CLIENT_URL=http://localhost:3000

# Supabase
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key

# JWT
JWT_ACCESS_SECRET=your-64-char-secret
JWT_REFRESH_SECRET=your-different-64-char-secret
JWT_ACCESS_EXPIRES_IN=15m
```

```
JWT_REFRESH_EXPIRES_IN=7d

# Email (Resend)
SMTP_HOST=smtp.resend.com
SMTP_PORT=465
SMTP_USER=resend
SMTP_PASS=your-resend-api-key
EMAIL_FROM=noreply@sumantravels.com

# Redis (Upstash)
REDIS_URL=your-redis-url
REDIS_TOKEN=your-redis-token

# Google Maps (Server-side)
GOOGLE_MAPS_API_KEY=your-maps-key

# Sentry
SENTRY_DSN=your-sentry-dsn
```

13. Development Milestones

Phase 1: Foundation (Week 1-2)

- Project setup (Next.js + Express + Supabase)
- Design system implementation (Tailwind config, colors, typography)
- Database schema creation
- Auth system (register, login, JWT)
- Header + Footer components

Phase 2: Core Features (Week 3-4)

- Home page (Hero, Services, Stats, Quick Booking)
- Fleet page with vehicle cards
- Vehicle detail view
- Pricing page with data table
- Contact form + API

Phase 3: Booking System (Week 5-6)

- Booking wizard (4 steps)
- Price calculation engine
- Booking API (CRUD)
- My Bookings page
- Booking confirmation emails

Phase 4: Admin & Polish (Week 7-8)

- Admin dashboard
- Vehicle management (CRUD)

- Driver management
- Booking management (status updates)
- Enquiry management
- SEO optimization
- Performance optimization
- Mobile responsiveness testing

Phase 5: Launch Prep (Week 9)

- Production deployment (Vercel + Render)
- Domain configuration
- SSL/HTTPS setup
- Final security audit
- User acceptance testing

14. Success Metrics

Metric	Target
Page Load Time	< 2 seconds
Lighthouse Score	> 90 (Performance, Accessibility, SEO)
Booking Conversion Rate	> 5%
Mobile Traffic	> 60% sessions
Uptime	99.9%
Customer Support Response	< 1 hour

15. Risk & Mitigation

Risk	Impact	Mitigation
Vehicle double-booking	High	Database unique constraints + transaction locks
Payment failures	High	Graceful error handling + manual payment option
Email delivery issues	Medium	Backup SMTP provider + SMS fallback
High traffic spikes	Medium	CDN + caching + auto-scaling
Data privacy concerns	High	GDPR compliance + data encryption + regular audits

16. Appendix

A. Vehicle Image Requirements

- Resolution: 1200x800px minimum

- Format: WebP with JPEG fallback
- Aspect ratio: 16:9
- Style: Professional, clean background, well-lit
- Angles: Front 3/4 view preferred

B. Content Guidelines

- Tone: Professional, premium, trustworthy
- Language: English (primary), Hindi (future)
- Keywords: “car rental Mumbai”, “airport transfer Mumbai”, “chauffeur service”, “premium car hire”

C. Legal Requirements

- Terms of Service page
- Privacy Policy page
- Cancellation Policy
- GST invoices for all bookings
- Driver background verification records

Document Owner: Product Team

Review Cycle: Monthly during active development, quarterly post-launch

Distribution: Frontend Team, Backend Team, Design Team, QA Team, Stakeholders

End of PRD — Suman Travels